

Precision spectroscopy of ultracold $^{86}\text{Sr}_2$ halo moleculesB. Iritani¹, J. Huang¹, W. Xu¹, G. Sim¹, R. Moszynski², and T. Zelevinsky^{1,*}¹Department of Physics, Columbia University, 538 West 120th Street, New York, New York 10027-5255, USA²Quantum Chemistry Laboratory, Department of Chemistry, University of Warsaw, Pasteura 1, 02-093 Warsaw, Poland

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We demonstrate the creation and field-free precision spectroscopy of ultracold $^{86}\text{Sr}_2$ molecules in the halo regime, where vibrational binding energies are dominated by long-range interactions. Using one-photon photoassociation with optimized wavefunction overlap, we produce stable samples of halo and neighboring weakly bound molecular states. We measure vibrational splittings of the three least-bound states with sub-100-Hz uncertainty, enabling a high-precision determination of absolute binding energies. These results establish $^{86}\text{Sr}_2$ halo molecules as a new platform for precision studies of long-range molecular forces, including quantum electrodynamics effects such as Casimir-Polder interactions, and provide a direct basis for isotope-shift spectroscopy sensitive to nanometer-scale mass-dependent forces.

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The rovibrational structure of cold diatomic molecules provides access to precision measurements that are complementary to, and in some cases surpass, those achievable with atoms, enabling new approaches to quantum metrology, quantum simulation, and tests of fundamental physics [1–6]. In particular, nonpolar alkaline-earth-metal dimers such as Sr_2 combine narrow optical and vibrational transitions with exceptional insensitivity to external fields, making them attractive systems for high-precision spectroscopy [1,7,8].

Weakly bound molecular states near the dissociation threshold can offer distinct advantages for precision studies, as their vibrational energies are dominated by long-range interactions and are especially sensitive to subtle modifications of the interatomic potential. In bosonic $^{86}\text{Sr}_2$, the existence of a halo state with an anomalously large spatial extent, arising from the exceptionally large field-free scattering length of ^{86}Sr , places the molecule in a universal regime where binding energies provide a direct probe of long-range quantum electrodynamics (QED) effects such as Casimir-Polder interactions [9–14]. While the halo-state binding energy has previously been inferred from photoassociative spectroscopy of atoms, precision spectroscopy of bound halo molecules themselves has remained experimentally unexplored.

Access to the full range of molecular bond lengths, including, importantly, this near-threshold regime, is highly relevant in the context of searches for physics beyond the standard model. Many extensions of known physics predict new mass-dependent, non-Newtonian interactions at nanometer length scales [15–22]. The vibrational spectrum of Sr_2 [23–25], governed by the interatomic potential, provides a sensitive

molecular-scale force probe, and comparison between high-precision measurements and *ab initio* quantum chemistry calculations enables constraints on such interactions [8]. Measurements of isotope shifts further suppress common-mode theoretical uncertainties, isolating isotope-dependent contributions to the molecular Hamiltonian [8].

Here, we report the creation and field-free precision spectroscopy of ultracold $^{86}\text{Sr}_2$ molecules in the halo regime. Using one-photon photoassociation (PA), we produce stable samples of halo and neighboring weakly bound vibrational states and identify optimal production pathways via photoassociative Autler-Townes spectroscopy. We measure the vibrational splittings of the three least-bound states with uncertainties below 100 Hz, allowing the determination of absolute binding energies for the $v = -2$ and $v = -3$ states. These measurements establish bound $^{86}\text{Sr}_2$ halo molecules as a new platform for precision studies of long-range molecular forces and provide a direct experimental basis for isotope-shift spectroscopy between $^{86}\text{Sr}_2$ and $^{88}\text{Sr}_2$.

An ultracold ^{86}Sr sample is prepared by cooling and trapping the atoms at $\sim 2\ \mu\text{K}$ from a Zeeman-slowed beam with a two-stage magneto-optical trap (MOT) on the broad $^1S_0 \rightarrow ^1P_1$ transition and the narrow $^1S_0 \rightarrow ^3P_1$ intercombination line. Subsequently, the atoms are transferred to a 914 nm one-dimensional optical lattice. After applying the spectroscopy pulse, the remaining atoms are absorption imaged on the $^1S_0 \rightarrow ^1P_1$ transition. The narrow-line MOT laser is stabilized to a high-finesse cavity and serves as the reference for phase locking the PA spectroscopy lasers [1,23].

Our PA method involves one-photon excitation of ground-state atoms into weakly bound vibrational states in the 0_u^+ or 1_u potential near the $^1S_0 \rightarrow ^3P_1$ atomic threshold [Fig. 1(a)]. The molecules then decay into several weakly bound states of the ground-state electronic potential, $X^1\Sigma_g^+$. The branching ratios for this decay are governed by the relative transition strengths. The near-threshold transition strengths are proportional to the vibrational wavefunction overlaps, or the Franck-Condon factors. A near-unity Franck-Condon factor ensures efficient

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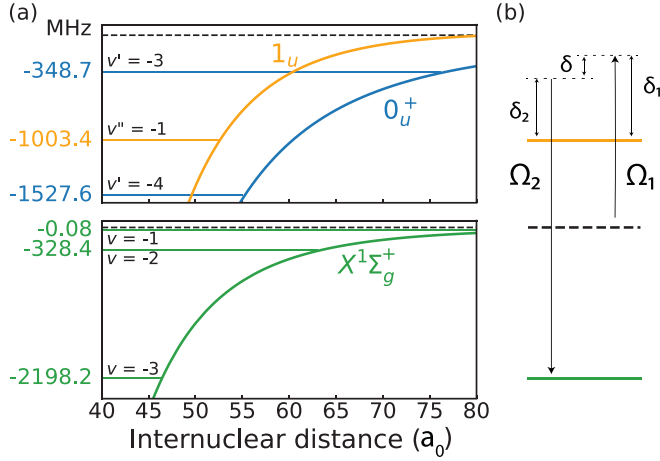


FIG. 1. Weakly bound $^{86}\text{Sr}_2$ molecules. (a) The ground-state electronic potential $X^1\Sigma_g^+$ asymptotes to the $^1S_0 + ^1S_0$ atomic threshold, and the excited-state potentials 1_u and 0_u^+ asymptote to the $^1S_0 + ^3P_1$ intercombination threshold. The weakly bound vibrational states are displayed (v, v' , and v'' for the three potentials, as labeled). (b) The Rabi frequency (Ω_1, Ω_2) and detuning ($\delta_1, \delta_2, \delta$) scheme for photoassociative Autler-Townes spectroscopy.

creation of molecules in a single vibrational state v of the ground state [23].

To measure the transition strengths, we use photoassociative Autler-Townes spectroscopy. In this two-photon process, the pump laser (Rabi frequency Ω_1) couples free atoms to the excited intermediate state, and the anti-Stokes laser (Rabi frequency Ω_2) connects the intermediate state to the target state. The anti-Stokes laser is kept at a fixed frequency, while the pump laser is scanned to reveal the characteristic Autler-Townes splitting of the photoassociation peak. The detuning configuration is shown in Fig. 1(b). For each measurement, we fit an Autler-Townes doublet of the form [26,27]

$$N(\delta_1, \delta_2, t) = N_0 \exp \left[-2\pi t \frac{\Gamma |\Omega_1|^2}{\Gamma^2 + 4\delta_1^2} \right] \times \left(1 - |\Omega_2|^2 \frac{|\Omega_2|^2 - 8(\delta_1 - \delta_2)\delta_1 + \Gamma_{\text{eff}}\Gamma(1 - 4\delta_1^2/\Gamma^2)}{|\Omega_2|^2 + |(\Gamma + 2i\delta_1)(\Gamma_{\text{eff}} + 2i(\delta_1 - \delta_2))|^2} \right), \quad (1)$$

where δ_1 and δ_2 are the detunings of the pump and anti-Stokes beams, respectively, and t is the probe time. Ω_1 and Ω_2 are the corresponding Rabi frequencies. Γ is the excited-state linewidth, dominated by power broadening, and Γ_{eff} is a decoherence rate that represents the relative linewidth between the pump and anti-Stokes lasers. Γ_{eff} is a free parameter that was fit to each scan, with an average value of 14(4) kHz.

If only the pump laser is present, i.e., the one-photon free-to-bound transition is driven, then we can set $\Omega_2 = \delta_2 = \Gamma_{\text{eff}} = 0$ and Eq. (1) becomes

$$N = N_0 \exp \left(\frac{-2\pi t |\Omega_1|^2 \Gamma}{\Gamma^2 + 4\delta_1^2} \right). \quad (2)$$

For each excited state, t is fixed and δ_1 is scanned to locate the resonance. Next, the laser is kept on resonance while t is varied, in which case Eq. (2) becomes

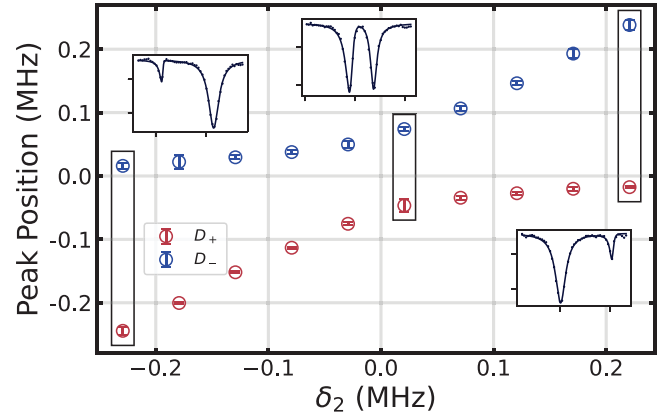


FIG. 2. Autler-Townes doublet peak positions for the $1_u(-1)$ and $X(-3)$ pair of states. Sample scans are shown in insets. The left peak in each scan is shown with blue circles in the main figure and the right peak is shown with red circles.

$N = N_0 \exp(-2\pi t |\Omega_1|^2 / \Gamma)$. The parameters Ω_1 and Γ are found by fitting these decay curves. Finally, with both lasers turned on, we obtain a series of Autler-Townes spectra as shown in Fig. 2.

From fitting the spectral lines to Eq. (1), we can directly determine Ω_2 and therefore the relative anti-Stokes transition strengths. Alternatively, the splitting between the higher- and lower-frequency Autler-Townes peaks ($D_{\pm}$) depends on δ_2 as

$$(D_+ - D_-)^2 = \delta_2^2 + |\Omega_2|^2. \quad (3)$$

Therefore, finding the minimum of this parabola as a function of δ_2 , measured with constant laser beam intensities, yields $|\Omega_2|^2$ and the relative transition strengths. Measuring the parabolas for several pairs of vibrational states yields a direct comparison of transition strengths. The results for all five measured state pairs with sizable transition strengths are displayed in Fig. 3 and Table I. The quoted transition strengths are normalized to the largest measured value.

Next, we proceed to creating and detecting molecules. Guided by the transition strength measurements summarized in Fig. 3, we photoassociate ground-state atoms to $0_u^+(-4, 1)$ and allow them to decay into the $X(-2, 0)$ state. An optional two-photon Raman π pulse is then applied via the $0_u^+(-2, 1)$ (-44.1 MHz) state to move the molecules to the halo state $v = -1$, and a resonance signal is observed as loss from $v = -2$. At the end of each sequence, the remaining $v = -2$ molecules are detected by first photodissociating to the $^1S_0 + ^3P_1$ threshold and then absorption imaging the atoms (Fig. 4).

TABLE I. Summary of measured relative transition strengths for photoassociation pathways. The values are normalized to the strongest transition.

$^1S_0 + ^3P_1$ levels	$^1S_0 + ^1S_0$ levels	
	$v = -2$	$v = -3$
$0_u^+(-3, 1)$	0.42(2)	—
$1_u(-1, 1)$	0.74(2)	0.57(2)
$0_u^+(-4, 1)$	1.00(4)	0.34(2)

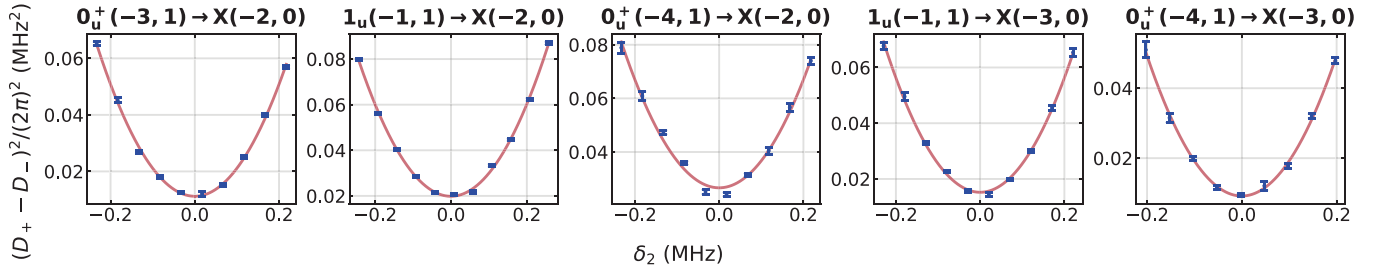


FIG. 3. Autler-Townes doublet separation parabolas. The minima of the parabolas are proportional to the relative transition strengths. The quantum numbers in parentheses correspond to (v, J) .

We can also directly photodissociate the halo state to the $^1S_0 + ^3P_1$ threshold. Attempts to directly populate the $v = -1$ halo state with one-photon photoassociation produced smaller molecule samples. The $1/e$ lifetimes of the $v = -2$ and the halo $v = -1$ molecule samples were measured to be ~ 50 ms, limited by two-body collisional loss and similar to timescales measured for $^{88}\text{Sr}_2$ molecules [28]. This indicates that the halo states do not suffer higher collisional losses than the more deeply bound molecules.

In order to characterize the ultralong-range portion of the molecular potential, we measure energy splittings between the three least-bound vibrational states. This is done using two-photon Raman spectroscopy of a trapped molecular sample, utilizing weakly bound states in the 0_u^+ potential as intermediate states. Typical line shapes are shown in Fig. 5. For each trace, the anti-Stokes laser frequency is fixed, and the pump laser is scanned across the resonance. The common detuning is much larger than any residual Zeeman splitting in 0_u^+ . The vibrational splitting is the difference between the

Raman laser frequencies. Both Raman lasers are phase locked to the 689 nm narrow-line MOT laser with different offsets. An additional offset between the Raman laser frequencies arises from double-passed acousto-optic modulators (AOMs) for both lasers. Radio frequencies for the offsets and AOMs are generated by direct digital synthesizers that are referenced to a commercial Rb frequency standard steered by a Global Positioning System (GPS) signal.

The resonance position for the $v = -2 \rightarrow v = -1$ transition is obtained by fitting a Lorentzian line shape [Fig. 5(a)]. Spectra for the $v = -2 \rightarrow v = -3$ transition are fit to a Rabi line shape due to laser-induced power broadening [Fig. 5(b)],

$$N = \frac{|\Omega|^2}{|\Omega|^2 + \delta^2} \sin^2\left(\frac{\sqrt{|\Omega|^2 + \delta^2}}{2}t\right), \quad (4)$$

where Ω and δ are the overall Rabi frequency and detuning, and t is pulse time.

By far, the dominant systematic effects are Stark shifts due to the probe and lattice lasers. We evaluate these systematics to <100 Hz uncertainty. For both transitions and for each contributing systematic effect, peak positions are measured at several laser intensities whose range is limited by laser power availability (Fig. 6). Five trials are used for each point, and a weighted linear fit is performed as a function of intensity. The shifts are then extrapolated to zero intensity. The systematic corrections and their uncertainties are reported in Table II.

Accounting for the systematic shifts, we report the vibrational splittings. We have measured the $v = -2 \rightarrow v = -1$ vibrational splitting to be $328.394 \pm 288(78)$ MHz and the

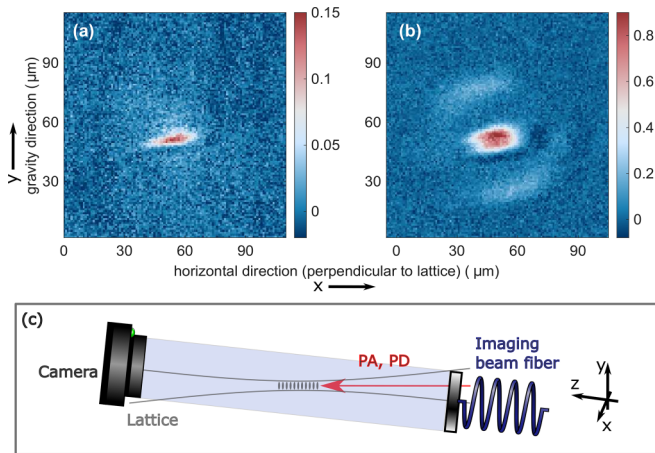


FIG. 4. Observation of $^{86}\text{Sr}_2$ molecule samples via photodissociation and subsequent time-of-flight ($150 \mu\text{s}$) absorption imaging of the resultant atomic fragments. Images are captured slightly off-axis from the optical lattice. The y axis is parallel to gravity and x axis is the horizontal direction perpendicular to the lattice axis. Color bars represent optical density. (a) The halo state, $v = -1$. Only the state with total angular momentum $J = 0$ is bound. (b) The $v = -2$ molecules, with $J = 0$ dissociation at the center and $J = 2$ fragments receiving a greater kinetic energy and forming the outer ring; odd values of J are forbidden by bosonic quantum statistics. (c) Illustration of the imaging geometry.

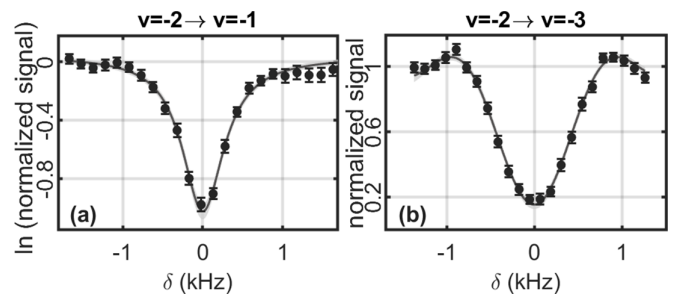


FIG. 5. Spectra of Raman transitions between weakly bound states at operational intensities for (a) $v = -2 \rightarrow v = -1$ and (b) $v = -2 \rightarrow v = -3$. Black points are (a) natural logarithm of normalized signal and (b) normalized signal vs δ . Fitted curves have (a) Lorentzian and (b) Rabi line shapes. The error bars display standard error of the mean.

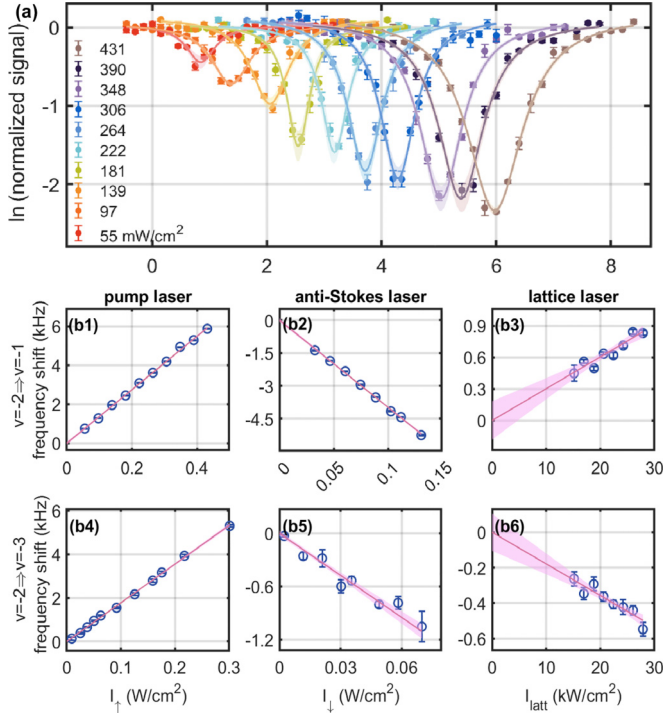


FIG. 6. (a) Sample spectra of a two-photon Raman transition exhibiting probe light shifts. Points represent data for ten different pump laser intensities. Error bars are statistical standard errors obtained from five trials. Lines represent Lorentzian fits of the natural logarithm of the detected atom number. Lower panels: Shifts induced by probe and lattice lasers vs light intensity for $v = -2 \rightarrow v = -1$ [panel (b), top row] and $v = -2 \rightarrow v = -3$ [panel (b), bottom row]. (b1), (b4) Light shifts induced by the pump laser; (b2), (b5) light shifts induced by the anti-Stokes laser; and (b3), (b6) light shifts induced by the lattice laser.

$v = -2 \rightarrow v = -3$ splitting to be 1869.681 045(47) MHz. The absolute binding energy of the $v = -1$ halo state was previously measured via photoassociative spectroscopy to be $-83.00(7)(20)$ kHz [11]. Therefore, we can report the absolute binding energies of the weakly bound states $v = -2$ and $v = -3$ (Table III).

The $^{86}\text{Sr}_2$ halo molecules realized in this work provide a distinctive platform for probing long-range QED effects in bound systems, in particular, Casimir-Polder (retardation) interactions. These molecules supply a series of bound vibrational states with widely varying spatial extent, enabling comparison between states with strongly differing sensitivities to long-range modifications of the interatomic potential.

TABLE II. Leading systematic shifts (light shifts due to the probe and lattice lasers) for $v = -2 \rightarrow v = -1$ and $v = -2 \rightarrow v = -3$ vibrational splitting measurements.

Stark shift (Hz)	$v = -2 \rightarrow v = -1$	$v = -2 \rightarrow v = -3$
Pump	1820(50)	742(32)
Anti-Stokes	-1855(46)	-147(25)
Lattice	567(37)	-335(23)
Total	532(77)	259(47)

TABLE III. Binding energies of weakly bound states in the $X^1\Sigma_g^+$ potential of $^{86}\text{Sr}_2$, calculated with the measured vibrational splittings and the halo-state binding energy previously determined via photoassociative spectroscopy [11]. The uncertainties for the $v = \{-2, -3\}$ states incorporate the uncertainty from the halo-state binding energy.

Vibrational state	Binding energy (kHz)
$v = -1$	$-83.00(7)(20)$ [11]
$v = -2$	$-328\,477.288(226)$
$v = -3$	$-2198\,158.333(231)$

In contrast to prototypical field-free halo systems such as $^4\text{He}_2$, which supports only a single bound state with a binding energy on the order of 10^7 Hz [29], $^{86}\text{Sr}_2$ is 2 orders of magnitude less deeply bound and offers access to multiple near-threshold as well as deeper states that can serve as internal references with reduced sensitivities to retardation effects (and heightened sensitivities to possible short-range new physics). Combined with ultracold temperatures and long molecular lifetimes, this enables high-resolution spectroscopy across a range of internuclear separations R .

To assess the sensitivity of the near-threshold spectrum to long-range effects [30–32], we model the interaction potential using standard dispersion coefficients C_6 and C_8 , with a retardation-induced modification to the C_6 term that becomes significant only for $R \gtrsim R_c \sim \lambda/(2\pi)$, where we use the wavelength $\lambda = 461$ nm for Sr. Solving the radial Schrödinger equation with the short-range phase constrained to reproduce the measured binding energies in the absence of retardation, we then introduce the long-range correction while fixing the short-range phase using the more deeply bound states. We find that the halo-state binding energy exhibits a strongly enhanced sensitivity to the long-range modification, with fractional shifts exceeding those of neighboring states by approximately 2 orders of magnitude. This differential response reflects the large spatial extent of the halo state and establishes near-threshold Sr_2 molecules as a sensitive probe of long-range interactions.

In conclusion, we have demonstrated the creation and field-free precision spectroscopy of ultracold $^{86}\text{Sr}_2$ molecules in the halo regime. By producing stable samples of the halo state and other weakly bound vibrational states, we measured the vibrational splittings of the three least-bound levels with sub-100-Hz uncertainty and determined their absolute binding energies. These measurements establish bound $^{86}\text{Sr}_2$ halo molecules as a sensitive platform for probing long-range molecular interactions, where relativistic and quantum electrodynamics effects are enhanced. Our results provide direct experimental input for high-precision *ab initio* quantum chemistry in a previously inaccessible regime and form a concrete basis for isotope-shift spectroscopy sensitive to nanometer-scale mass-dependent forces.

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Data availability. The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing,

depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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